

**Proposed Data Centre
Development at No. 2-10,
Tai Yuen Street, Kwai
Chung, N.T.**

Technical Specification for
Structural Steel

Revision: 0

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Structural Steel

1.0 Design

1.1 General

1.1.1 Design Standards

Ensure compliance of the design with the requirements of Code of Practice for the Structural Use of Steel 2011 by BD for all building and temporary works.

1.1.2 Alternative Design

The Contractor may propose members or assemblies designed on an experimental basis as an alternative to members shown on the Drawings. Evidence must be produced to show that at least three similar members or assemblies, selected at random, have been subjected to a Strength Test to confirm their capacity in accordance with:

- a) BS 5950 Part 1 Clause 7.3.4, or equivalent, for hot rolled sections.
- b) BS 5950 Part 5 Section 10.6.4, or equivalent, for cold rolled sections.

Where a production run of the member or assembly is carried out evidence of consistency of production must be produced by carrying out Check Tests in accordance with:

- a) BS 5950 Part 1 Section 7.3.6, or equivalent, for hot rolled sections.
- b) BS 5950 Part 5 Section 10.6.6, or equivalent, for cold rolled sections.

1.2 Fabrication and Erection Documents

1.2.1 Erection Documents

Prepare and submit to the Architect at least six weeks before starting to erect steelwork, details of the following:

- a. Method of erection;
- b. Type of craneage;
- c. Calculations of erection stresses;
- d. Temporary guys and bracing for use during the erection.

Do not commence erection or deviate from the Drawings prior to receiving Approval

1.2.2 Design Responsibility

Employ a Registered Structural Engineer to take responsibility for the design, erection and construction of all temporary works and falsework. No consent or dissent of the Architect will relieve the contractor of his sole responsibility for the falsework design, construction and obtaining Approval.

1.2.3 Welding Symbols

Show welding on the drawings using welding symbols to BS 499: Part 2.

1.2.4 Drawings Responsibility

Accept responsibility for any errors in fabrication drawings subsequently found whether or not Approval to the drawings has been obtained.

1.2.5 Record Drawings

On completion of the erection, revise or redraw fabrication drawings to show any agreed modifications made during the course of the Contract. Submit one copy negative or velograph of each drawing to the Architect for record purposes.

2 Materials

2.1 General

2.1.1 References to Standards

Where the Contractor proposes to substitute any other standard or code of practice either in its entirety or in part, for any of those specified submit details of such together with two complete copies for Approval. Generally, Approval will only be given where the Architect considers the proposals to give at least an equivalent standard of the finished work.

2.1.2 Materials and Workmanship

Ensure that all structural steelwork is in accordance with Code of Practice for the Structural Use of Steel 2011 for building and temporary works.

2.1.3 Approvals

Prior to commencing construction submit for Approval any details required of the welding plant, techniques and procedures, jigs, workshop facilities and construction devices e.g. cranes, scaffolding etc., which the Contractor proposes using for the fabrication and erection of the steelwork. The information must be in sufficient detail to enable the Architect to Approve or otherwise.

2.1.4 Safety

Comply with safety recommendation contained in BS EN 1011-1: 1998 and BS EN 1011-2:2001 and with the Factories and Industrial Undertakings Ordinance throughout the fabrication, transport, handling and erection of structural steelwork.

2.1.5 Method of Manufacture

Obtain and submit for Approval details of the method to be used for steel manufacture. Do not use:

- a) Steel made by the Bessemer process;
- b) Rimming steel.

2.1.6 Steel Grades

Structural steelwork to comprise weldable structural steel in accordance with the following standards as appropriate:

- a) Hot rolled sections (excluding hollow sections) and plates: BS EN 10025:2004 - Grade S275, S275JR, S355JR, S355JO;
- b) Hot finished hollow sections: BS EN 10210 - Grade S275JOH, S355JOH;
- c) Cold formed hollow sections: BS EN 10219-1:1997.
- d) Cold rolled galvanized sections: either: BS 2989 Grade Z35, coating type G600, surface finish N, or: Hot dipped coated steel sheets to BS EN 10147 Grade FE E3509, coating type G600, surface finish N.

2.1.7 Hot Rolled Sections

Dimensions, mass, tolerances and rolling margins for:

- a) Universal beams, columns, joists, tees and channels to BS 4: Part 1;
- b) Hollow sections to BS EN 10210-2:1997.
- c) Angles to BS EN 10056-1:1999.

2.1.8 Cold Rolled Sections

Dimension, mass and tolerances of cold rolled sections to BS EN 10162:2003. Do not use cold rolled sections without Approval.

2.1.9 Cold Rolled Hollow Sections

Dimension, mass and tolerances of cold rolled hollow sections to BS EN 10219:1997. Do not use as substitute for hot finished hollow sections.

2.1.10 Plates, Flats and Bars

- a) For plates dimensions and mass tolerances to BS EN 10029;
- b) For wide flats dimensions and mass tolerances to either BS 4360 or Euronorm 91 Class II thicknesses tolerances;
- c) For bars dimensions and mass tolerances to either BS 4360 or Euronorm 59 - Square bars, Euronorm 60 - Round

Ensure tolerances on thickness is wholly over that shown on the drawings unless otherwise specified.

2.1.11 Surface Quality

Surface quality of steel to be in accordance with the following:

- a) Sections: either BS 4360 or BS EN 10163 Part 3 Class C;
- b) Plates and wide flats: either BS 4360 or BS EN 10163 Part 2 Class A.

2.1.12 Surface Defects and Repair

Steel with surface defects within the limits of the following may be repaired and used:

- a) Sections: either BS 4360 or BS EN 10163 Part 3 Class C;
- b) Plates and wide flats: either BS 4360 or BS EN 10163 Part 2 Class A.

Obtain Approval to carry out surface defect repairs by welding.

2.1.13 Surface Rusting

Use only steel with Grade A, B or C surface rusting in accordance with BS 7079.

2.1.14 Maximum Carbon Equivalent

Steel used in built-up welded assemblies or at welded connections must have a carbon equivalent value not exceeding that appropriate to the grade of steel given in Table 2 of BS 5135 where this may be required for the avoidance of hydrogen cracking in welds unless the Contractor's proposed welding procedures are such to make this requirement unnecessary.

2.1.15 Marking Steelwork

Clearly mark all steelwork for identification and erection purposes as follows:

- a) Identification: ensure that all steel elements are identified in accordance with BS EN 10025 as applicable, indicating grade, cast number and manufacturer's identification mark. Where indicated on the Drawings as being unmarked, tag steelwork and leave free of all other marking or hard stamping;
- b) Grade S355 steel: in addition to marking in accordance with BS EN 10025 mark steel with a continuous pale blue water paint line. Mark immediately, with light blue water painting any part cut from S355 steel which does not have a light blue marking;
- c) Cold work steel: mark cold worked areas of member with orange water paint;

- d) Marking for erection purposes: tag, hard stamp or pencil weld each member with an individual identification mark, which cannot be obliterated by surface coatings. Prepare a marking drawing identifying markings and indicating position of member in structure;
- e) Permanent identification: supply and fix to structure after erection, a brass identification plate. Plate to indicate with lettering not less than 15 mm high, etched or punched to a depth not less than 1.5 mm, the following:
 - i. Steelwork Contractor's name;
 - ii. Grade of steel in structure;
 - iii. Date of fabrication.

2.1.16 Non-Available Materials

Obtain Approval for any equivalent alternative required when the specified material is not available

2.1.17 Manufacturers Certificates

Provide the manufacturer's certificates for all steel sections, rivets, bolts, screws, nuts and washers delivered to Site.

2.2 Bolts, Accessories, Connections and Fittings

2.2.1 Steel Grade for Connections and Fittings

Use Grade 8275 steel as a minimum, subject to design suitability, unless otherwise specified.

2.2.2 High Strength Bolts and Nuts

- a) To BS 3692: Grade 8.8;
- b) To the strength of BS 3692: Grade 8.8 and the tolerances of BS 4190;
- c) To BS 4395: Part 1 used as ordinary high strength bolts in accordance with BS 4604: Part 1 and Code of Practice for the Structural Use of Steel 2005, without the hardened steel washers.

2.2.3 BS 4190 Holding Down Bolts and Nuts

Grade 4.6 but purpose made with either normal bolt head or screwed both ends with two extra nuts.

2.2.4 BS 3692 Holding Down Bolts and Nuts

Grade 8.8 but purpose made with either normal bolt head and screwed both ends with two extra nuts.

2.2.5 Black Bolts and Nuts

To BS 4190: Grade 4.6.

2.2.6 Drive Screws

Use self-tapping metallic drive screws to BS 4174.

2.2.7 Plain Washers For Black Bolts

Round washers to be to BS 4320, Form A made from cold rolled strip in the hardened condition to BS 1449: Part 1: Section 1.15, Grade CS4 chamfered on one outside edge.

2.2.8 Spring Washers for Black Bolts

Washers to be single coil square or rectangular steel spring washers to BS 4464.

2.2.9 Tapered Washers

Made from cold formed strip in the hard condition to BS 1449: Part 1: Section 1.15, Grade CS4 to similar dimensions and tolerances as shown in BS 4395: Part 1.

2.2.10 Bolt Lengths

Ensure threaded portion of each bolt projects through the nut by at least one thread.

2.2.11 High Strength Friction Grip Bolts

Use bolts and associated nuts and washers to BS 4395: Part 1 unless otherwise specified.

2.2.12 Rivets

Use mild steel rivets for general engineering purposes to BS 4620.

2.2.13 Turnbuckles Or Similar

Obtain Approval for specially made elements and components such as turnbuckles.

2.2.14 Marking Bolts and Accessories

Clearly mark all bolts, screws, nuts and washers by either embossed or indented markings identifying the relevant British Standard.

2.3 Welding Consumables

2.3.1 Quality of Electrodes, Rods and Wires

Select to give weld metal properties not less than the minimum specified for the grade of steel being welded.

2.3.2 Consumables for Welding Steels to BS EN 10025 and BS 4360

To BS EN 1011-1:1998 & BS EN 1011-2:2001.

2.3.3 Electrodes For Welding Grade S355 Steels

To be capable of depositing no more than 15ml of diffusible hydrogen per 100g of deposited weld metal, as defined in BS 639.

2.3.4 Covered Electrodes For Manual Metal Arc Welding

To BS 639.

2.3.5 Filler Wires For Gas-Shielded Welding

To BS EN 440:1995.

2.3.6 Cored Electrodes

To BS EN 758:1997.

2.3.7 Filler Wires and Fluxes for Submerged ARC Welding

To BS EN 756:2004.

2.3.8 Continuous Covered Electrodes

Of an Approved manufacture.

2.3.9 Manufacturer's Test Certificates

Provide Architect with relevant manufacturer's certificates for each batch of electrodes/rods/coils of wire.

2.3.10 Storage of Electrodes

Keep electrodes in unbroken packets in accordance with BS EN 1011-1:1998 & BS EN 1011-2:2001. Any drying or baking of consumables before issue shall be carried out in accordance with the manufacturers recommendations.

3 Workmanships

3.1 Transportation and Storage

3.1.1 General

Make all necessary arrangements for transport of steel from the fabrication yard to the site.

3.1.2 Handling

Carefully load, stack, and unload to avoid damage of any description Replace damaged items at Contractor's own expense unless the Architect permits rectification of damage.

3.1.3 Special Handling

- a) Where necessary; use, special slings or lifting points or both for handling, loading and unloading in fabrication shops, during transit and on Site;
- b) Provide special supports, packings and lashings on vehicles, trucks and on ship (if appropriate) to prevent chafing;
- c) Ensure small components receive special handling and stacking;
- d) Ensure unpainted components, e.g. high strength friction grip bolts and associated nuts and washers are adequately protected against corrosion;

3.1.4 Bolts, Nuts, Washers, And Screws:

- a) Transport and store in sealed or wire clipped bags labelled to identify:
 - i. Contents;
 - ii. Number;
 - iii. Grade;
 - iv. Standard;
 - v. Manufacturer;
 - vi. Batch or cast number.
 - vii. Store in a dry area dedicated to the purpose.

- viii. Do not mix contents of opened bags.

3.1.5 Exposure to Pollutants

- a) Take precautions to minimise exposure of steelwork to atmospheric or chemical pollution before and after fabrication;
- b) Store steelwork awaiting erection clear of the ground and keep different members separate. Lay or stack to avoid accumulation of water or dirt on or against any of the surfaces. Provide suitable packings between layers of stacked steelwork. Ventilate covered steelwork sufficiently to preclude condensation;
- c) Store and stack steel members in such a manner that markings are clearly visible.

3.1.6 Road Transport

- a) Use vehicles long enough to carry components with permitted overhang at either end;
- b) Provide intermediate supports to which a component can be tied with strops to prevent oscillation;
- c) Sit components on softwood supports covered with soft clean pads;
- d) Ensure buckles to strops do not touch components.

3.1.7 Sea Transport

- a) Containerise steelwork destined to be shipped overseas;
- b) Do not ship steelwork as deck cargo unless in containers;
- c) Ensure containers can be handled at ports of exit and entry if vehicle cannot be ferried;
- d) Pack containers so that no components touch. Use softwood with soft padding and man-made fibre ropes and strops to prevent movement and contract;
- e) Bag fasteners and accessories;
- f) Bundle small components and prevent contact;
- g) Chromate dip galvanized steel, which is not to be overcoated, when it is to be containerised for longer than three weeks or when going to go through the tropics.

3.1.8 Repair of Damage Incurred in Transport

- a) Inspect steelwork after transportation;
- b) Set aside components with damage to steel or coatings;
- c) Report damaged steel to the Architect;
- d) If damaged component is not rejected by the Architect repair by agreed method;
- e) Repair coatings as specified immediately.

3.2 Fabrication

3.2.1 Access

Provide access for the Architect at all reasonable times to all places where work is being carried out, with all necessary facilities for inspection during fabrication and off- site assembly.

3.2.2 Geometric Form

Keep all material, before and after fabrication, straight unless required to be of curvilinear form. Keep free from twists. Avoid excessive stack-up of sections to prevent damage and distortion.

3.2.3 Identification

Identify at all stages of fabrication.

3.2.4 Surface Preparation

Before fabrication remove all oil and grease by use of a suitable solvent from all bare surfaces and remove all mill scale, rust and dirt by either:

- a) Shot blasting to surface finish designation Sa 2 to BS 7079:Part A1;
- b) Mechanical wire brushing to surface finish designation St 2 to BS 7079:Part A1

3.2.5 Post Cleaning

After cleaning adopt one of the following methods:

- a) Apply a coat of blast primer, not greater than 20 microns OFT. Where subsequent welding is to be carried out ensure that a weld through blast primer is used which will not give off noxious fumes during welding. Ensure compatibility with final protective system; or
- b) Fabricate steelwork within four hours of cleaning and apply protective system primer; or
- c) Reclean steel after fabrication by shot blasting or mechanical wire brushing to standard required for protective coating and apply protective system primer.

3.2.6 Execution

Carry out all fabrication to Code of Practice for the Structural Use of Steel 2005.

Conform with the requirements for accuracy of construction with respect to levels, dimensions, alignment and fabrication tolerances of members and joints given in the above British Standards

3.2.7 Setting Out

Set out all holes and cuts along the length of a member from one datum line perpendicular to its longitudinal axis. Mark centres of all holes at standard back marks/cross centre unless otherwise shown on the Drawings.

3.2.8 Temporary Attachments

Design and fabricate all temporary attachments for use as assembly/erection aids to ensure their use will not damage the structure or its surface protection.

3.2.9 End Plates

- a) For flexible end plate connections:
 - i. To be not less than 12 mm thick;
 - ii. To extend across, but not be connected to the top flange;
- b) For sealing hollow sections;
 - i. Suspend suitable desiccant (Silica-gel) in bags in each void prior to sealing up;
 - ii. Weld sealing plate over end.

3.2.10 Jigs

Use jigs for repetitive, interchangeable or complicated work.

3.2.11 Measurement and Setting Out Marks

Make all measurements by steel tape or bands calibrated at 20°C and used at the correct tension.

3.2.12 Continuous Structures

Make allowance in setting out continuous structures for dimensional variations due to temperature variations.

3.2.13 Straightening Bending And Cambering Of Steel Sections

Submit proposals to the Architect where steel sections are required to be straightened, bent to a specified radius or precambered and methods to be adopted to perform these operations, for his approval in each case. Ensure that the adopted method does not weaken or deface the member.

3.2.14 Post Fabrication Making Good

Clean and prepare all mechanical damage, weld burns and welded surfaces as follows:

- a) By shot blasting; or
- b) Mechanical or hand wire brushing;
- c) All bare and welded surfaces must also be made good and primed;
- d) Plates over 25 mm that are to be connected by high strength friction grip bolts and found to be bowed or rippled must be machined flat before the steel leaves the fabrication works.

3.2.15 Cutting Steel

- a) Cut by sawing, shearing, cropping, nibbling, laser or machine flame cutting, unless otherwise instructed. Do not use manual gas cutting without Approval;
- b) Shearing must not distort steelwork;
- c) Saw cut slightly oversize to allow for machining and/or grinding;
- d) Flame cut oversize where necessary to allow for removal of notches by machining.

3.2.16 Cut Edges

- a) Remove burrs, sharp arrises and slag from edges of all cut members;
- b) Dress machine sheared or cropped edges to a neat finish, free from distortion.

3.2.17 Machining Generally

- a) Grind stiffeners, plates and the like to fit the profile of the parent member with diagonal cuts to clear the root radius;
- b) Machine stanchion splices and butt joints of compression members true and square;
- c) Splay cut projecting corners of cleats, plates, etc. and machine the edges and ends of all flanges and end plates of girders or other members except where Universal beams or flats with reasonably square edges are used;

- d) Machine the edges of all web and flange plates of welded girders square unless specified otherwise. Where plates are specified to be planed, cut to sufficient size to permit 3 mm to be planed off all edges. The edges of floor plates, gutter plates, and trough flooring may be sheared;
- e) Machine square all butt ends of compression members and those in close contact throughout so that applied loads are evenly transmitted over the entire area of the section.

3.2.18 Flame Cutting

Grind off heat affected metal or notches where gas cutting is used. Where gas cutting is applied to S275 steel over 25 mm thick or to S355 steel of all thickness, machine any gas cut edge which is to be subject to fluctuating stresses along its length but not subsequently incorporated in a weld so as to remove not less than 3 mm of the flame cut profile.

3.2.19 Forming Holes for Bolts and Rivets

Form holes by any of the following methods:

- a) Drilling full size;
- b) For plate less than 15 mm thick: punching 2 mm less in diameter than the required size and subsequently reaming out to the required diameter;
- c) Punching full size and meeting the following conditions:
 - i. The punching operations must not in the opinion of the Architect unduly distort the material;
 - ii. The holes must be free from burrs which would prevent solid seating of the parts when tightened;
 - iii. The thickness of the material through which the hole is punched must not exceed the diameter of the hole;
 - iv. The thickness of the material must not be greater than 12 mm unless Approved;
 - v. In spliced connections the holes in the mating surfaces must be punched from the same direction.
- d) Holes must not to be punched full size in any of the following conditions:
 - i. In tension flanges at locations where plastic hinges have been assumed in the design analysis;
 - ii. In elements of rigid connections in which local yield lines have been assumed;
 - iii. Where repetition of loading makes fatigue critical in the member design.
 - iv. Comply with clause 5.3.7 of BS 5950: Part 1 that requires locations where any of these conditions apply are shown on the Drawings.

3.2.20 Built-Up Sections

- a) Drill after the members have been assembled and tightly clamped together;
- b) Where not practicable, drill the holes to a diameter at least 2 mm less than the required size and ream after assembly to the full diameter.

3.2.21 Removing Burrs

Remove all burrs from holes from assembly. Parts which have been drilled to receive black bolts need not be separated to remove burrs. Parts drilled to receive HSFG bolts to be separated and the surfaces cleaned of all burrs and drilling fluid to ensure the coefficient of friction is not compromised.

3.2.22 Holes for Black Bolts

Finished holes for black bolts must not be more than 2 mm in diameter larger than the diameter of the bolt passing through them for bolt diameters up to 24 mm, and not more than 3 mm larger than the diameter of the bolts for bolt diameters over 24 mm, unless otherwise specified or required by the design.

3.2.23 Holes for High Strength Friction Grip Bolts

Use hole sizes specified in HOLES FOR BLACK BOLT where the number of plies in the grip does not exceed three, if the number of plies exceed three, form the holes in the two outer plies as before and the diameter of the hole in the inner plies not more than 3 mm larger than that of the bolt.

3.2.24 Fitted Bolts, Pins Or Rivets

- a) Holes for rivets:
 - i. All matching holes for rivets to register with each other so that a gauge 2 mm less in diameter than the diameter of hole will pass freely through the assembled members in a direction at right angles to such members;
 - ii. Finished holes to be more than 2 mm in diameter larger than the diameter of the rivet passing through them, for rivet diameters up to 24 mm, and not more than 3 mm greater than the diameter of the rivet for rivet diameters over 24 mm, unless otherwise specified;
 - iii. Holes for rivets must not be formed by a gas cutting process;
- b) Holes for fitted bolts:
 - i. Drill holes for fitted bolts to a diameter equal to the nominal diameter of the shank subject to a tolerance of +0.15 mm and -0 mm;
 - ii. Use tacking bolts or cramps on parts connected together with fitted bolts. Drill through all the thicknesses at one operation and ream to size;
 - iii. All holes not drilled through a thicknesses at one operation are to be drilled to a smaller size and reamed out after assembly. Where this is not practicable, drill and ream the parts separately through hard bushed steel jigs.

3.2.25 Slotted Holes

Drill first then ream out smooth to required size.

3.2.26 Prevent Moisture Ingress

Make provision to prevent moisture ingress to sealed hollow members where these are holed for bolts.

3.3 Mechanical Jointing

3.3.1 Surface Preparation

Prepare and prime the contact surfaces at riveted, bolted and screwed joints. If the contact surfaces are specified to be coated or otherwise treated, e.g. machined surface finish, obtain Approval to the joint before assembly.

3.3.2 Use Of High Strength Friction Grip Bolts

- a) Clean the interface of members jointed with high strength friction grip bolts to the specified surface preparation and leave unpainted;
- b) Use high strength friction grip bolts in accordance with BS 4604:Part 1. Tighten the bolts and nuts by the torque-control method using a calibrated power-operated or a hand-operated torque wrench to achieve the proof loads set out below:

Nominal size	Proof Load, kN (minimum shank tension)
M12	49.4
M16	92.1
M20	144
M22	177
M24	207
M27	234
M30	286

- c) Use tapered washers on surfaces with a slope $>3^\circ$.

3.3.3 Wrench Calibration

- a) Maintain the wrench in a proper working condition in accordance with the manufacturer's recommendations. Calibrate the wrench regularly using a calibrated bolt load meter and using a bolt from the batch of bolts to be used in the works as follows:
- At the commencement of each new working day or shift;
 - With each change of bolt diameter;
 - With each change of bolt grip length in excess of 1/5th of the length used in the calibration of the wrench in accordance with BS 4604: Part 1.
- b) Re-apply the wrench to bolts previously tightened in a multi-bolt connection to ensure all bolts have the correct torque;
- c) Ensure that the bolt used in the calibration is, as far as is practicable, representative of bolts to be used in the works. No additional lubrication should be applied unless it is to be applied to bolts incorporated in the permanent works.

3.3.4 Calibration Of Bolt Load Meter

The bolt load meter for measuring bolt shank tension must be calibrated by an Approved laboratory before tightening of bolts and nuts commences and at regular intervals agreed by the Architect. During re-calibration, a replacement calibrated bolt load meter must be provided on the Site. Submit calibration results to the Architect at least one week before the bolt load meter is used.

3.3.5 Black Or Precision Bolted Connections

- Keep all bolts, nuts and washers free from rust and dip in linseed oil immediately before connections;
- Provide each bolt with at least one steel washer placed under the nut or bolt head, whichever element is to be rotated during the tightening operation;
- Tighten all bolt heads and nuts against surfaces normal to the bolt axes. Use tapered washers as necessary to meet this requirement. Where bolted connections are subject to vibration provide the nuts with lock nuts or castle nuts and split pins.

3.3.6 Alignment, Temporary Connections and Permanent Fixings

- Do not distort holes or members during alignment;
- Do not enlarge holes without prior Approval;
- Drifts used during the fabrication or erection of the steelwork must be of such shape and dimensions as to locate the work accurately. Drifts must not cause any damage to or deformation of the component parts, nor alter in any way the sizes of the holes provided for the permanent bolts;

- d) Temporary bolts used for the initial or trial assembly of any part of the steelwork must have the same hole clearances as the permanent bolts;
- e) Tighten all bolts securely in the finished work so that the length of each bolt after tightening projects at least one thread beyond the other face of the secured nut.

3.3.7 Bolted Connections with Provision for Sliding

Provide slotted hole through one member and a round hole in the other. Connect with shouldered bolts, spring washers, flat washers, castle nuts and splits pins. Form round holes with a diameter less than the width of the slot and place the shoulder of the bolt against the faying surface of the unslotted member enabling the bolt to be tightened on this member. Locate the spring washer under the head of the bolt and the flat washer under the castle nut.

3.4 Welding

3.4.1 General

- a) Prior to fabrication of bridgework carry out welding and flame cutting trials using representative samples of materials to be used in the work in accordance with BS 5400: Part 6;
- b) Provide an experienced and competent operative to supervise welding. Submit evidence that all welders are capable;
- c) When directed by the Architect, keep a site record that identifies the welders responsible for major welds.

3.4.2 Use Of Welding Equipment and Accessories

- a) Use welding plant, equipment and accessories that comply with the requirements of BS 638;
- b) Use individual transformers or motor generator that are fitted with a current density control and situated close to the welder so that he may adjust the current;
- c) Store and use welding consumables in accordance with BS 5135, BS 639 and the manufacturers recommendations;
- d) Store electrodes in dry conditions and where necessary keep in heated quivers while in use;
- e) Use covered electrodes to BS 639;
- f) Regularly calibrate, using an ammeter, the transformers or motor generators to ensure that current is kept within the range specified by the manufacturers for electrodes;
- g) Do not open packets except for sampling or when electrodes are to be use;
- h) Reject electrodes with broken, damp or damaged coatings.

3.4.3 Preparation for Welding:

- a) Submit in writing, for Approval, details of proposed welding procedure for each type of welded connection;
- b) Keep the surfaces to be welded free from all scale, grease, paint, rust and other matter. Prepare all parts to be welded in a manner that ensures a close fit.
- c) Prepare fusion surfaces to BS 5135. Dry surfaces and warm as necessary, to remove any condensation.

3.4.4 Additional Requirements for Site Welding

- a) Surfaces to be welded must be clean, free from paint, grease, contaminants dry and free from condensation. Where the steel is damp or the temperature is low then warming with a low

temperature flame may be used. Thermal indicating crayons may be used to determine the plate temperature and the area pre- heated;

- b) Protection from the weather: provide adequate protection to shield the welding from the weather. Tent type structures made from fireproof and waterproof material may be used to shield the weldment from the wind and rain. Such temporary structures to be mounted on stable scaffolding supports, and a stable platform to be provided for the welder. Where welding to hollow sections is required ensure draughts are prevented from blowing through the section;
- c) Electrodes: bake on site at a temperature between 100°C and 150°C and where necessary store in heated quivers for use at the welders convenience;
- d) Power supply: may be from a transformer or generator to provide a three phase power supply of the required voltage. Locate the generator or transformer as close as possible to the welder to prevent a drop in voltage along the cables length. Ensure all the cables are as short as possible, routed away from the path of construction vehicles, and adequately protected;
- e) Earth run: adequately earth all electrical plant. Ensure the welding return from the weldment is adequate in cross section and correctly connected and earthed.
- f) Weld detail: ensure the welded joint is of the correct profile and fit up. As a general guide do not carry out welding where the gap is less than 1 mm or more than 3 mm unless a backing strip is provided. Where a backing strip is provided the weld gap is not to exceed the greater of 5 mm or the thickness of the material being welded. Adequately support all welded joints with props or restraints that ensure relative movement of the fusion faces is prevented;
- g) Work in confined spaces: provide adequate ventilation and formulate plans of evacuation and escape before the work commences.

3.4.5 Method of Welding

- a) Carry out welding in accordance with the requirements of BS 5135. Manual metal-arc weld unless otherwise specified;
- b) Execute welding in a flat or horizontal position wherever possible. Do not weld overhead without Approval;
- c) Ensure electrodes used in conjunction with fluxes give a weld deposit having mechanical properties at least equal to the minimum specified for the parent metal. Use electrodes and fluxes strictly in accordance with the manufacturer's recommendations.

3.4.6 Removal of Slag

Remove welding slag by chipping before depositing subsequent runs, and thoroughly wire brush the surfaces.

3.4.7 Tack Welds

Do not use tack welding unless approved. Tack welds to be minimum 50 mm long. Form location welds, where Approved, of the same quality and size as the first run of the main weld which when completed, fuse completely with the ends of the location welds and form a reasonably regular finished profile. Do not weld temporary attachments to principle joints. Obtain Approval of the position of welds for temporary attachments.

3.4.8 Weld Defects

- a) Welds will be rejected if they contain:
 - i. Serious undercut or overlap;
 - ii. Lack of penetration;
 - iii. Lack of inter-run fusion;

- iv. Slag inclusions;
- v. Worm holes and porosity.
- b) Any limitations on these defects will be as stated in BS 5135. Do not exceed the allowable defects for weld quality category C as stated in BS 5135, Table 18 for butt welds and Table 19 for fillet welds.
- c) Take particular care when welding hollow sections, plates and angles to avoid distortion by using correct sequences and speeds.

3.4.9 Fillet Welds

Deposit fillet welds to the required length, throat thicknesses and with partial or full penetration as specified.

3.4.10 Butt Welds

- a) Form butt welds as full penetration welds between prepared fusion faces, unless otherwise directed;
- b) Carry out back chipping, grinding or gouging of the deposited weld as required to obviate imperfections in the root run such that the root edges or root faces are not out of alignment by more than $1/8$ of the thickness of the thinner material for material up to 12 mm thick or by more than 1.5 mm for thicker material;
- c) Grind butt welds flush without loss of parent metal.
- d) In the fabrication of built-up assemblies ensure, all butt welds in each component are completed, wherever possible, before the final assembly

3.4.11 Stud Shear Connectors

Weld stud shear connectors by automatic stud welding gun strictly in accordance with the manufacturer's recommendations unless specified otherwise.

3.5 Erection of Steelwork

3.5.1 Erection

- a) Carry out all erection to:
 - i. BS 5950 : Part 2, or equivalent, for hot rolled sections in building;
 - ii. BS 5950 : Part 7, or equivalent, for cold rolled sections in building;
 - iii. BS 5400: Part 6, or equivalent, for plates and sections in bridgework.
- b) Conform with the requirements for accuracy of construction in respect of levels, dimensions and alignment given in the above British Standards and those contained in the "Schedule of Tolerances" in Appendix H to this Specification.

3.5.2 Erectors

Carry out all work employing trained and experienced erectors. Provide evidence of the erector's competence when directed by the Architect.

3.5.3 Safety

Comply with the relevant regulations on safety including the 'Factories and Industrial Undertakings Ordinance', the recommendations of BS 5531 as applicable and:

- a) Use only recently tested equipment for lifting and remove from site any damaged equipment;

- b) Do not carry out complex operations at a height except from safe platforms large enough to take operators and equipment. Maintain platforms and safe access until work has been accepted and any added protection completed.

3.5.4 Stability

Ensure stability of structure at all stages of construction.

3.5.5 Sequence of Operations

Prepare and submit details of the proposed method of erection for the Architect's consent. Details submitted must include type of plant and equipment to be used and if necessary drawings and calculations of any temporary work. The giving of consent by the Architect will not in any way relieve the Contractor of his responsibility for safe erection of permanent work or the safe erection and subsequent dismantling of temporary work.

3.5.6 Plant and Equipment

Do not employ any plant or equipment which is, or in the Architect's opinion likely to be, unsuitable, unsafe, or likely to cause damage to the structure during erection or damage to existing buildings etc.

3.5.7 Prevention of Distortion

Do not distort steelwork or exceed the stresses specified in BS 5950: Part 1, or equivalent.

3.5.8 Approved Sequence

Erect structural steel according to the sequence agreed with the Architect.

3.5.9 Falsework

When specified, design, construct and dismantle falsework to BS 5975. Supply all staging and all labour, tools, erection plant, lifts, services, nuts and washers and other necessary materials. Provide temporary bracings and guys to ensure adequate resistance to wind and stability against collapse during construction.

- a) Submit records of inspection of the completed formwork and falsework to the Architect for information before carrying out any further works that may affect the stability of the formwork and falsework. Such records are to be certified by the Quality Control Engineer.
- b) Monitor falsework levels and make any necessary adjustments for settlement, or other deflection or movements.

3.5.10 Setting Out, Levelling and Location

Set out, level and position:

- a) Using steel tapes or bands for setting out that are calibrated at 20°C and used at the correct tension;
- b) To a level that is levelled to a datum point and aligned to a base line during erection. Agree both datum point and setting out lines before commencement;
- c) That ensures all steelwork is accurately and securely located to the correct profile and camber, if any, before assembly;
- d) That ensures the structural steelwork is not subjected to stresses in excess of those for which it was designed.

3.5.11 Erection of Steelwork

- a) Do not erect fabricated steelwork until all welded joints and specified protective coatings have been inspected, tested where required, and approved;
- b) Align structural members accurately to the specified tolerances and obtain Approval before any permanent welding or tightening of bolts is carried out;
- c) Do not bed stanchion bases or grout anchor bolts until the steelwork has been plumbed, levelled, aligned and approved.

3.5.12 Steelwork by Sub-Contractor

- a) Take responsibility for checking all the work prepared by others that may be incorporated into the work or the dimensions of any structure affecting steel erection. As a result of the check report any discrepancies or shortfalls to the Architect immediately;
- b) Ensure that the Sub-Contractor supplies any items required to be cast into concrete. Cast in such items using templates provided by the Sub-Contractor where these are necessary.

3.6 Cleaning, Surface Preparation and Protection (Post Fabrication)

3.6.1 Cleaning and Coating in the Suppliers Workshop

Ensure that the supplier provides sufficient covered and enclosed workshop space to enable blast cleaning, priming coat and undercoat painting to be carried out in close sequence. The blast cleaning section of the workshop must be fitted with extractors and filters to prevent and deposition of dust on metal surfaces to be painted.

3.6.2 Cleaning Surfaces of Bare Metal

Thoroughly clean down to remove all dirt, grease and the like, using a suitable solvent and thoroughly chip, scrape, wire-brush flame clean or grit blast to remove all loose scale back to clean metal.

3.6.3 Contaminated Surfaces

- a) Weld spatter: clean of all surfaces, keep paint films free of embedded foreign metallic particles;
- b) Concrete deposits etc.: wash or clean off immediately they occur and if necessary make the surfaces good to the satisfaction of the Architect;
- c) Oil or grease: clean with white spirit.

3.6.4 Painted and Other Coated Surfaces

Unless otherwise directed by the Architect clean the final workshop applied coating to the external surfaces on site by:

- a) Washing with a solution of an Approved liquid detergent followed by rinsing with clean fresh water and allowed to dry thoroughly before overcoating;
- b) Thoroughly cleaning rivet heads, bolts heads and the surfaces surrounding them;
- c) Removing all loose particles, dust and debris immediately prior to application of paint.

3.6.5 Atmospheric Corrosion

Do not apply protective coating to surfaces bearing atmospheric corrosion products or other chemicals which may be harmful to the coating or succeeding coating. Clean off these contaminants, which may include any remaining after surface preparation as specified in this work section and any produced by welding, to an Approved extent that is compatible with the specified protective system.

3.6.6 Blast Cleaning

- a) Carry out blast cleaning in accordance with BS 7079: Part A1 and that produces surface qualities as follows:
 - i. For surfaces to be galvanized to 140 microns: Sa 2 (prior to pickling)
 - ii. Note: Surfaces to be galvanized to 85 microns do not require blast cleaning.
 - iii. For surfaces to be metal sprayed: as specified in BS 2569.
 - iv. For surfaces to be painted: Sa 2%.
- b) Non-metallic abrasive should be free from contamination and recovered material is to be cleaned to the satisfaction of the Architect before re-use;
- c) Do not exceed a maximum amplitude (peak to trough) of the blast cleaned surface of 0.1 mm;
- d) Clean blasted surfaces by brush or vacuum. Do not touch by hand. Do not contaminate in any other way.
- e) Apply protective coating within 4 hours of blast cleaning;
- f) Before commencing any follow on operations submit, for Approval, a sample blast cleaned steel panel measuring not less than 150 mm x 150 mm x 6 mm thick adequately protected by sealed clean polythene wrapping. The Approved sample will be retained by the Architect for comparison with the prepared steelwork.

3.6.7 Mechanical Cleaning

Carry out mechanical cleaning to surface quality designation St 2 of BS 7079 by power driven tools, such as carborundum grinding discs, chipping hammers and needle guns, followed by steel-wire brushing and dusting to remove all loosened material. Avoid excessive brushing of the metal through prolonged application to rotary wire brushes. Protect surfaces within 48 hours of mechanical cleaning.

3.6.8 Pickling

Carry out pickling of steel by the "Footner" process as specified in BS 5493. Apply the first priming coat of paint as soon as the steel has dried and is still warm.

3.6.9 Flame Cleaning

Plan and control the sequence of operations for flame cleaning steelwork to avoid the risk of distortion and buckling. Clean the flame cleaned surfaces of all loosened material. Apply priming coat while the steel is still warm. Control the heat input so that no harmful distortion of the steelwork occurs.

3.6.10 New Galvanized Surfaces:

Thoroughly clean to remove all dirt, oil and grease and pre-treat with either an etching primer or Approved etching compound.

3.6.11 Previously Painted Galvanized Surfaces

Thoroughly degrease previously painted galvanized surfaces. Where corrosion has spread under the paint film, remove old paint with an organic solvent type paint remover.

3.6.12 Cleaned or Coated Members

- a) Ensure that, unless specified otherwise, exposure times for blast cleaned wire brushed or coated surfaces (other than at joints) are not longer than those given in the following table:

Surface	Enclosed Workshop	Outside
a. Blast cleaned	4 hours	Nil
b. Metal Sprayed	4 hours	Nil
c. Mechanically Cleaned	48 hours	Nil
d. Blast Primer on 'a'	Maximum exposure of weeks with up to weeks of this time outside	
e. Etch primer on 'b'	24 hours	Nil
f. Primer on 'c'	48 hours	Nil
g. 1st Undercoat on 'd', 'e' or 'f'	48 hours	Nil
h. Subsequent Undercoats	As agreed by the Architect	

Note: 'Outside' refers to any area outside a fully enclosed workshop.

- b) The tabulated exposure times refer to any part of the surface being blast or mechanically cleaned, metal sprayed or painted;
- c) For surface condition 'd' where blast primed steelwork is exposed outside, the minimum dry paint film thickness over the peaks of the blast cleaned steel must not be less than 13 microns.
- d) Zinc rich coated steelwork may only be exposed outside for the minimum period to allow movement at the fabricator's works. Such movement to be under the supervision of the Architect or his representative and unless the weather conditions are favourable the sections are to be covered.
- e) Where it is necessary to overcoat two pack paints of the epoxide or polyurethane type, re-coat within 48 hours of application. Where it is not possible to overcoat within this time the paint film to be abraded to produce a roughened surface, given a flash coat of suitable primer, and allowed to dry for at least 4 hours before application of the next coat of the system.
- f) No steelwork to be loaded for transport until the paint system has been Approved as being sufficiently dry for handling.
- g) Keep steelwork to be site painted clean and dry. Where necessary provide temporary shelters to simulate, as far as is practicable, workshops conditions.

3.7 Application of Metal Coatings

3.7.1 Hot Dip Galvanizing

- a) To BS 729: minimum thickness of zinc coating of 85 microns unless otherwise specified;
- b) Apply metal coatings to BS 729 only to components of a tensile strength up to and including that of general grade high strength friction grip bolts unless otherwise specified;
- c) Make good small areas of galvanized coating damaged by welding, cutting or by rough treatment during transit or erection either by the use of low melting point zinc alloy repair rods, powders made specifically for this purpose, or by the use of at least two coats of zinc rich paint to BS 4652. Apply sufficient material to provide a zinc coating at least equal in thickness to the galvanized coating.

3.7.2 Sprayed Metal Coating

To BS 2569: Part 1. The nominal thickness of coating to be 100 microns. Apply one flood coat of polyurethane (or similar approved) sealer immediately after spraying.

3.7.3 Electroplating

To BS 3382: Parts 1 and 2: Coating thickness: 5 microns minimum.

3.7.4 Chromium Plating

To BS 4641. Ensure that the protection offered is at least the same as that provided for the adjacent main structural steel element.

3.7.5 Metal Coating On Part Only of Assembled Section

Apply the metal coating before the remainder receives its priming coat.

3.7.6 Protection of Hsfg Bolted Joints

- a) Zinc or aluminium sprayed steelwork: where the metal spray is carried over the interfaces of the HSFG bolted joints, take the etch primer on the parent metal between 10 mm and 20 mm inside the perimeter of the joint area. The outer surfaces and edges of the joint material may also be given a coat of the same primer in order to reduce surface preparation required after the joint has been made;
- b) Painted steelwork: where metal spray is specified only at the interfaces of HSFG bolted joints, carry the metal spray between 10 mm and 20 mm outside the perimeter of the joint area and the blast primer taken between 10 mm and 20 mm inside the perimeter. If the outer surfaces and edges of the joint materials are metal sprayed, these may be given a coat of etch primer in order to reduce subsequent surface preparation;
- c) Painted steelwork: where the interfaces of HSFG bolted joints are bare steel, take the blast primer between 10 mm and 20 mm inside the perimeter of the joint area. The outer edges and surfaces of the joint material may also be given a coat of blast primer in order to reduce subsequent surfaces preparation.

3.7.7 Protection of other Bolted Joints

- a) Shop joints apply metal spray plus etch primer or blast primer alone to parent and joint material. Assemble the joint immediately after the first undercoat of the parent system has been applied to the contact surfaces;
- b) Site joints: apply to all surfaces, excepting those of fasteners, the same painting system in the shop as is to be applied to the parent metal surfaces, unless otherwise specified.

3.7.8 Protection of Non-Bolted Joints

- a) Riveted joints: shop and site joints: apply metal spray plus etch primer or blast primer alone, to parent and joint material. Assemble the joint immediately after the first undercoat of the parent system has been applied to the contact surfaces;
- b) Welded joints: unless otherwise specified, treat welds and surfaces which have been affected by welding with the protective system which is applied to the parent surfaces.

3.7.9 Surfaces Adjacent to Treated Joints

- a) Keep metal spray at least 5 mm clear of areas to be welded and mask off these areas during spraying;
- b) Where paints other than blast primers are to be applied to the parent surfaces before the making of a joint they must be stepped back at 30 mm intervals commencing at 80 mm from welded joints and at 10 mm from the perimeter of all other joints.

3.7.10 Painting and Sealing of Joints

- a) As soon as possible after joints have been Approved, paint the parent and joint material, exposed parts of bolts, nuts and washers, weld and affected areas to bring them up to the same state of preparation and painting as the adjoining surfaces;
- b) Seal all bolted joints against the ingress of water. Before painting commences and, subject to Approval, plug gaps at joints with an Approved filler. Seal the perimeter of all joints with subsequent coats of paint.

3.7.11 Paint for Steelwork Protection Systems

- a) In accordance with BS 5493, Section 2, Table 4 unless specified otherwise. Organic zinc-rich paint to comply with BS 4652. Do not use lead-based paint for finishing coats;
- b) Supplied in sealed containers of not more than 5 litres capacity. Each container to show the following information:
 - i. The name of the manufacturer;
 - ii. The paint manufacturer's reference number;
 - iii. Intended purposes, type of pigment and binder;
 - iv. Batch number, date of manufacture, expiry date and pot life, and
 - v. Colour, gloss, drying times and flash point.

3.8 Testing

3.8.1 General Requirements

3.8.1.1 General

Allow sufficient time in the Works programme for the testing specified below. Do not incorporate materials until relevant tests have been carried out and Approval obtained.

3.8.1.2 Testing by Manufacturer

Carry out inspection and testing on BS EN 10025 products by cast and documentation supplied in accordance with Euronorm 21. Specific testing and documentation to include chemical analysis; tensile tests and impact tests.

3.8.1.3 Testing of Bolts, Nuts and Washers

- a) Provide test specimens and carry out testing of bolts and associated nuts and washers in accordance with the relevant BS in an Approved Laboratory employed and paid for directly by the Contractor, when so directed by the Architect;
- b) Rate of sampling to be one bolt per identifiable batch or 1% whichever is the lesser unless otherwise specified.

3.8.1.4 Test Certificates

Obtain a manufacturer's certificate of test in accordance with the appropriate standard for each steel batch relating to steel to be used in the works. Documentation to include chemical analysis impact test and tensile test.

3.8.1.5 Testing of Sections

- a) Provide one test specimen for every 40 tonnes or part thereof of each section of same thickness from the same cast. For the purpose of this clause "same thickness" means similar sections with a variation in thickness not exceeding +5 mm. Test specimens to be taken from sections selected at random on Site by the Architect;
- b) Prepare the test specimens to BS 4360 or BS EN 10025 as directed and appropriately mark and deliver them to a Direct Testing Contractor employed and paid direct by the Authority for testing. Testing and documentation to include tensile tests and impact tests.

3.8.1.6 Failure of Tests

If any test specimens do not comply with the relevant BS, the Architect may require further tests at the Contractor's expense or reject the whole or part of the particular consignment, which must be removed from Site forthwith.

3.8.1.7 Ultrasonic Testing

Test quality grade steel plates and flats which may, as a result of welding be subject to lamellar tearing, using ultrasonic testing using equipment as described in Section 2 of BS 5996 and, unless otherwise specified or agreed, be of quality LC3 as defined in Section 3 of that document.

3.8.1.8 Ability of Welders

Submit evidence that all welders are capable of passing the training tests to BS 1295 and if specified, either:

- a) For welds where welding procedure approval is required to BS 4870: Carry out approval testing of welders to BS 4871 : Part 1 for steel and stainless steel and BS 4871 : Part 2 for aluminium as appropriate; or
- b) For welds where welding procedure approval is required to BS EN 288: Carry out approval testing of welders to BS EN 287 : Part 1 for steel and stainless steel and BS EN 287 : Part 2 for aluminium as appropriate;
- c) For welds where welding procedure approval is not required: Carry out approval testing of welders to BS 4872: Part 1 for steel and stainless steel and BS 4872: Part 2 for aluminium as appropriate.

3.8.1.9 Particulars of Inspecting Authority

Submit to the Architect the name of the proposed inspecting authority endorsing welder certificates and records of approval tests for welding procedures. The name to be submitted at least 3 weeks before approval tests start or, if approval tests are not required, at the same time as the welder certificates are submitted.

3.8.1.10 Weld Procedure Testing to British Standards

Unless acceptable documentation relating to the welding of similar connections is available, carry out approval testing of each welding procedure in accordance with, either:

- a) BS 4870: Part 1 for steel and BS 4870: Part 2 for aluminium as appropriate; or
- b) BS EN 288: Part 3 for steel and BS EN 288: Part 4 for aluminium.

3.8.1.11 Additional Destructive Weld Tests as Selected by the Architect

Prepare specimen welds and subject to one or more of the following tests as required:

- a) Butt welds:
 - i. Transverse tensile test to BS 18 and BS 4360 or BS EN 10025 as appropriate;
 - ii. Transverse bend tests to BS 1639 with the root of the weld in tension and compression respectively;

- iii. Side bend tests;
- iv. Macro-examinations;
- b) Fillet welds
- c) Carry out examination not less than 16 hours after the completion of most welds, the exception being butt welds thicker than 40 mm and any welding in grade 55 material which should not be examined sooner than 40 hours after completion;
- d) If any test specimens do not comply with the relevant BS, the Architect may require further tests at the Contractor's expense or reject the whole or part of the particular consignment, which must be removed from Site forthwith.

Strictly follow the procedures of welding established in the successful Approval test for the Works.
Carry out further Approval tests if either the material or procedure changes.

3.8.1.12 Additional Non-Destructive Testing Of Steel, Stainless Steel And Aluminium

Arrange for a percentage of each welder work to be selected by the Architect and subjected to non-destructive testing as follows:

- a) All welds - 100% visual inspection in accordance with BS 5289;
- b) Butt welds - 10% ultrasonic in accordance with BS 3923: Part 1 level 28; or 10% magnetic particle inspection in accordance with BS 6072 where the steel is less than 8 mm thick. (Not for stainless steel and aluminium);
 - Or 10% dye penetrant testing in accordance with BS 6443;
- c) Fillet welds - 10% Magnetic particle inspection in accordance with BS 6072. (Not for stainless steel and aluminium);
 - Or 10% dye penetrant testing in accordance with BS 6443;
 - Or 10% ultrasonic in accordance with BS 3923 : Parts 1 level 28;
- d) Examination is to be carried out not less than 16 hours after the completion of most welds, the exception being butt welds thicker than 40 mm and any welding in grade 55 material which should not be examined sooner than 40 hours after completion;
- e) Where any weld is judged to be unsatisfactory by the Architect a further 10% of that welders work is to be tested. If further defects are found then all weld in the group should be tested by the same method.

3.8.1.13 Acceptance Criteria

- a) Reject welds if they contain:
 - i. Serious undercut or overlap;
 - ii. Lack of penetration;
 - iii. Lack of inter-run fusion;
 - iv. Slag inclusions;
 - v. Worm holes and porosity.
- b) Limitations on these defects are defined in BS 5135, as category C as stated in BS 5135, Table 18 for butt welds and Table 19 for fillet welds.

3.8.1.14 Weld Repair

Cut out and repair all defective welds in accordance with BS 5135.

3.8.1.15 Engagement of Specialist Testing Contractor

- a) Carry out tests required to satisfy the requirements of BS 4870, BS 4871, BS 4872, BS EN 287 and BS EN 288 by an Approved Laboratory employed and paid for directly by the Contractor.

- b) Carry out tests required by the Architect by a Direct Testing Contractor employed and paid for directly by the Authority.

3.8.2 Testing Of Paints

3.8.2.1 Sampling and Testing

Permit authorized sampling of materials on site. Set aside unopened sealed containers of materials for testing when directed by the Architect.

3.8.2.2 Paint Samples

Sample each batch of paints in accordance with BS 3900: Part A1.

Testing Paints

Carry out the following tests on each sample:

BS 3900: Part C2	Surface drying test;
BS 3900: Part C3:	Through dry test;
BS 3900:Part E1	Bend test
BS 3900:Part E2	Scratch test;
BS 3900:Part E7	Resistance;
BS 3900:Part F3	Resistance to artificial weathering;
BS 3900:Part F4	Resistance to continuous salt spray;
BS 3900:Part F8	Resistance to humid atmospheres containing sulphur dioxides.

3.8.2.3 Testing Laboratory

- a) Tests for paints to be carried out by an Approved Laboratory employed and paid for directly by the Contractor;
- b) Test certificates to be submitted to the Architect before painting starts;
- c) Where manufacturers test certificates are available these may be accepted in lieu of a retest at the Architect's discretion.

3.8.2.4 Gloss Finish

Prepare a sample panel to indicate the standard of gloss finish and obtain Approval. Ensure application matches agreed sample.

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